PYTHON FOR DATA SCIENCE · LESSON 9 OF 29

Functions: Reusable Code, Arguments, Scope and Docstrings

Package logic into reusable functions. Learn def, parameters and arguments (positional, keyword, default, *args, **kwargs), return values, scope, docstrings and type hints, plus testing small functions on real data.

Python class 7 | Python 3.13 | Beginner | 100 min

By the end of this lesson you can

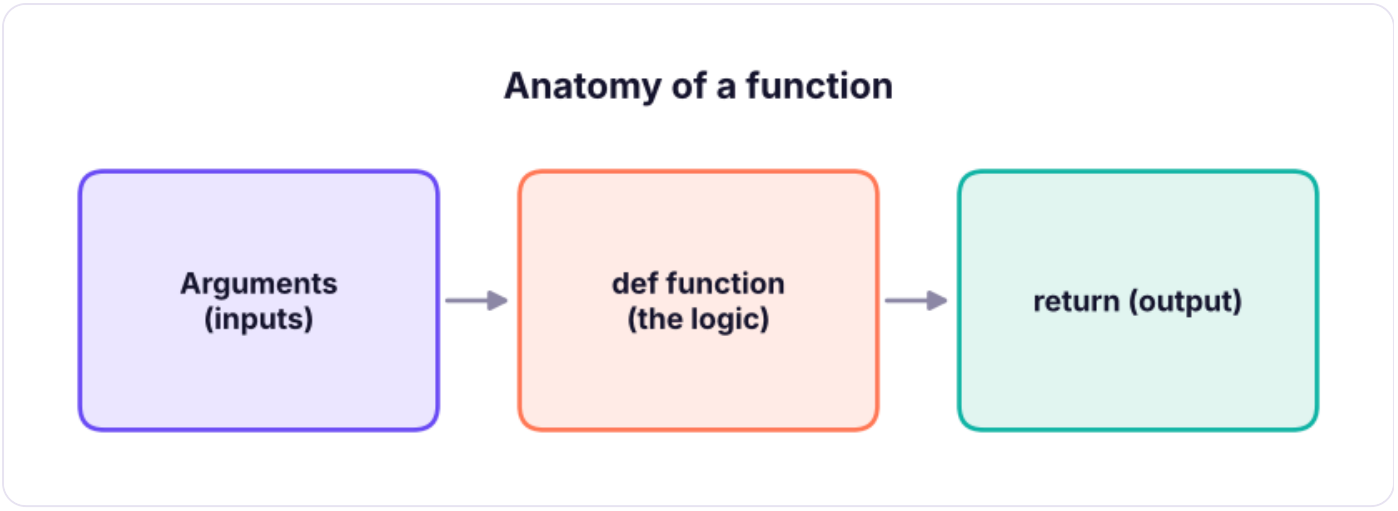
- Define and call functions with positional, keyword and default arguments
- Use *args and **kwargs and return multiple values
- Explain local vs global scope and avoid the mutable-default trap
- Write docstrings and type hints that make functions self-explaining
- Build small, testable functions for data cleaning and metrics

Datasets used

`superstore_sales.csv`

Why functions?

A **function** is a named, reusable block of code. It removes copy-paste, makes code testable, and lets you give a name to an idea (`clean_phone`, `margin_pct`).



A function takes inputs (arguments), runs its body, and gives back an output (return value).

Python

```
def greet(name):
    return f"Hello, {name}!"
print(greet("Asha"), greet("Ravi"))
```

OUTPUT

```
Hello, Asha! Hello, Ravi!
```

Parameters and arguments

Python

```
def margin_pct(profit, sales, decimals=1):
    """Profit as a percentage of sales."""
    return round(100 * profit / sales, decimals)

print(margin_pct(25, 200))                    # positional
print(margin_pct(sales=200, profit=25))       # keyword: order does not matter
print(margin_pct(25, 200, decimals=3))        # default overridden
```

OUTPUT

```
12.5
12.5
12.5
```

Kind	Example	Notes
Positional	`f(1, 2)`	Matched by order
Keyword	`f(b=2, a=1)`	Matched by name; clearer
Default	`def f(a, b=10)`	Optional argument; defaults go last
`*args`	`def f(*args)`	Extra positional arguments as a tuple
`**kwargs`	`def f(**kwargs)`	Extra keyword arguments as a dict
Keyword-only	`def f(a, *, b)`	`b` must be named

Python

```
def total(*nums): return sum(nums)
def describe(**info): return ", ".join(f"{k}={v}" for k, v in info.items())
def report(title, *values, sep=" | ", **opts): return title + ": " + sep.join(map(str,
values)) + f" {opts}"
print(total(1, 2, 3), total(), describe(city="Pune", year=2025))
print(report("Sales", 10, 20, 30, sep=",", currency="INR"))
nums = [4, 5, 6]; opts = {"decimals": 2}
print(total(*nums), margin_pct(7, 30, **opts))      # unpack a list / dict into arguments
```

OUTPUT

```
6 0 city=Pune, year=2025
Sales: 10,20,30 {'currency': 'INR'}
15 23.33
```

Return values

Python

```
def stats(values):
    n = len(values)
    mean = sum(values) / n
    var = sum((v - mean) ** 2 for v in values) / (n - 1)
    return min(values), max(values), round(mean, 2), round(var ** 0.5, 2)     # a tuple
lo, hi, mean, sd = stats([12, 15, 11, 18, 14])
print(lo, hi, mean, sd)

def say(): print("no return")
print(say())                                  # a function with no return gives None
```

OUTPUT

```
11 18 14.0 2.74
no return
None
```

> **TIP** **Return, do not print**
>
> A function that **prints** can only be read by humans. One that **returns** can be tested, combined and reused. Print at the very end, outside the function.

Scope: local vs global

Python

```python
rate = 0.18                       # global
def price_with_gst(price):
    gst = price * rate            # reads the global
    return price + gst            # gst is local: gone after the call
print(price_with_gst(1000))
try: print(gst)
except NameError as e: print("NameError:", e)

counter = 0
def bump():
    global counter                # needed to *rebind* a global (rarely a good idea)
    counter += 1
bump(); bump(); print(counter)
```

OUTPUT

```
1180.0
NameError: name 'gst' is not defined
2
```

Rule of thumb (LEGB): Python looks in **L**ocal, then **E**nclosing, then **G**lobal, then **B**uilt-in scope. Prefer passing values in and returning values out.

The mutable-default trap

Python

```
def add_item(x, bucket=[]):          # BAD: one list shared by every call
    bucket.append(x); return bucket
print(add_item(1), add_item(2))

def add_item_ok(x, bucket=None):     # GOOD
    if bucket is None: bucket = []
    bucket.append(x); return bucket
print(add_item_ok(1), add_item_ok(2))
```

OUTPUT

```
[1, 2] [1, 2]
[1] [2]
```

Lists, dicts and sets as default values are created **once**, when `def` runs. Use `None` and create the object inside.

Docstrings and type hints

Python

```
def clean_phone(raw: str, country_code: str = "+91") -> str:
    """Return a phone number as '+91 XXXXX XXXXX', or '' if it is not 10 digits.

    >>> clean_phone("98765-43210")
    '+91 98765 43210'
    """
    digits = "".join(ch for ch in raw if ch.isdigit())[-10:]
    if len(digits) != 10: return ""
    return f"{country_code} {digits[:5]} {digits[5:]}"

print(clean_phone("98765-43210"), repr(clean_phone("12345")), clean_phone("+91 (98765) 43210"))
help(clean_phone)
```

OUTPUT

```
+91 98765 43210 '' +91 98765 43210
Help on function clean_phone in module __main__:

clean_phone(raw: str, country_code: str = '+91') -> str
    Return a phone number as '+91 XXXXX XXXXX', or '' if it is not 10 digits.
```

```
>>> clean_phone("98765-43210")
'+91 98765 43210'
```

Type hints do not change how the code runs, but editors and tools such as `mypy` use them, and they document intent. Write the docstring first line as a short summary; tools like `help()` and IDE hovers display it.

Functions are objects

Python

```
def square(x): return x * x
fn = square                                  # assign, store in a list, pass as an
argument
def apply_twice(f, x): return f(f(x))
print(fn(4), apply_twice(square, 3), [f.__name__ for f in (square, len, str)])

def make_multiplier(n):                      # a function that returns a function (a
closure)
    def times(x): return x * n
    return times
double, triple = make_multiplier(2), make_multiplier(3)
print(double(7), triple(7))
```

OUTPUT

```
16 81 ['square', 'len', 'str']
14 21
```

Functions on real data

Small, pure functions are easy to test and to reuse across `apply`, comprehensions and pipelines.

Python

```python
import pandas as pd
s = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"])

def discount_band(d: float) -> str:
    if d == 0: return "none"
    if d <= 0.10: return "low"
    if d <= 0.20: return "medium"
    return "high"

def summarise(df: pd.DataFrame, by: str) -> pd.DataFrame:
    """Sales, profit and margin % per group."""
    g = df.groupby(by).agg(sales=("sales", "sum"), profit=("profit", "sum"), lines=
("sales", "size"))
    g["margin_pct"] = (100 * g.profit / g.sales).round(1)
    return g.round(0).astype({"sales": int, "profit": int})

s["band"] = s["discount"].map(discount_band)
print(summarise(s, "band").sort_values("margin_pct", ascending=False))
print(summarise(s, "region").sort_values("sales", ascending=False))
```

OUTPUT

```
            sales    profit  lines  margin_pct
band
none    123936290  21718963   3480        18.0
low      21864636   1286023    726         6.0
medium   35864461   -588882   1217        -2.0
high     19748182  -4063940    900       -21.0
           sales   profit  lines  margin_pct
region
South   53952768  4702012   1656         9.0
West    51958520  4719833   1614         9.0
North   49018161  4555830   1530         9.0
East    46484119  4374490   1523         9.0
```

The same `summarise` works for any column. That reuse is the point: write once, call with `by="category"`, `by="segment"`, and so on.

Testing a function

Python

```python
assert margin_pct(25, 200) == 12.5
assert clean_phone("12345") == ""
assert discount_band(0.15) == "medium"
print("all checks passed")
```

OUTPUT

```
all checks passed
```

`assert` is the quickest test. Larger projects use `pytest` (Lesson 14).

Recursion in one paragraph

A function can call itself; it needs a **base case** that stops the calls.

Python

```python
def factorial(n):
    return 1 if n <= 1 else n * factorial(n - 1)
def fib(n, memo={}):                    # deliberate cache; see the lesson on functools.lru_cache
    if n < 2: return n
    if n not in memo: memo[n] = fib(n - 1) + fib(n - 2)
    return memo[n]
print(factorial(6), [fib(i) for i in range(12)], fib(60))
```

OUTPUT

```
720 [0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89] 1548008755920
```

Python limits recursion depth (about 1,000 calls), so prefer loops for deep or large problems.

Common mistakes

Mistake	Fix
Forgetting `return` (function gives `None`)	Check the value you expect back
Mutable default arguments	Use `None` and create inside
Calling `f` instead of `f()`	`f` is the function object; `f()` runs it
Modifying a global inside a function without `global`	`UnboundLocalError`; pass and return values instead
Function does too many things	One job per function; compose small ones
Positional order confusion	Use keyword arguments for clarity

Summary

KEY TAKEAWAYS

- `def name(params): ... return value` defines a function; no `return` means it returns `None`.
- Arguments can be positional, keyword, default, `*args` (tuple) and `**kwargs` (dict); put defaults last.
- Variables created inside a function are local; read globals freely, but prefer parameters and return values.
- Never use a mutable object as a default; use `None`.
- Docstrings and type hints document intent; `assert` gives quick tests.
- Functions are objects: you can pass them around, store them and return them (closures).

Quick quiz

6 questions · tap an answer to check it

1. What does a function without a `return` statement return?

A 0

B None

C An empty string

D An error

Answer: B. Python returns `None` implicitly.

2. What is wrong with `def f(x, items=[]):` ?

- **A** Nothing
- **B** The default list is created once and shared across calls
- **C** Lists cannot be arguments
- **D** It must be a tuple

Answer: B. Use `items=None` and create the list inside.

3. What does `*args` collect?

- **A** Keyword arguments in a dict
- **B** Extra positional arguments in a tuple
- **C** Only the first argument
- **D** Nothing

Answer: B. `**kwargs` collects the keyword ones.

4. Which scope does Python search first?

- **A** Global
- **B** Built-in
- **C** Local
- **D** Enclosing only

Answer: C. LEGB: Local, Enclosing, Global, Built-in.

5. How do you call `f(a, b)` with a list `pair = [1, 2]`?

A `f(pair)`

B `f(*pair)`

C `f(**pair)`

D `f[pair]`

Answer: B. `*` unpacks the list into positional arguments.

6. Why return values instead of printing them?

A Printing is slower

B Returned values can be tested, reused and combined

C Print is not allowed in functions

D Python requires it

Answer: B. Printing is for the final display only.

EXERCISE

Function problems

(1) Write `is_prime(n)`. (2) Write `percent_change(old, new)` returning a rounded percentage and guard against `old == 0`. (3) Write `top_n(df, col, n=5)` that returns the top-n Superstore rows by a column. (4) Write a function that accepts any number of numbers and returns their average, or `None` if none are given. (5) Add `assert` tests for each.

Show solution (worked answer)

Python

```
def is_prime(n):
    return n > 1 and all(n % d for d in range(2, int(n ** 0.5) + 1))
def percent_change(old, new): return None if old == 0 else round(100 * (new - old) /
old, 1)
def top_n(df, col, n=5): return df.nlargest(n, col)
def average(*nums): return sum(nums) / len(nums) if nums else None
assert [x for x in range(20) if is_prime(x)] == [2, 3, 5, 7, 11, 13, 17, 19]
assert percent_change(0, 5) is None and percent_change(200, 250) == 25.0
assert average() is None and average(2, 4) == 3
print(top_n(s, "sales", 3)[["order_id", "sales"]])
```

INTERVIEW CORNER

Questions you may be asked

Q: Difference between an argument and a parameter? Parameters are the names in the `def`; arguments are the values you pass when calling.

Q: What are `*args` and `**kwargs`? They let a function accept any number of extra positional and keyword arguments; the first arrives as a tuple, the second as a dict.

Q: What is a closure? A function that remembers variables from the scope in which it was created, even after that scope has finished.

Q: Why is a mutable default argument dangerous? It is evaluated once at definition time, so state leaks between calls.

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